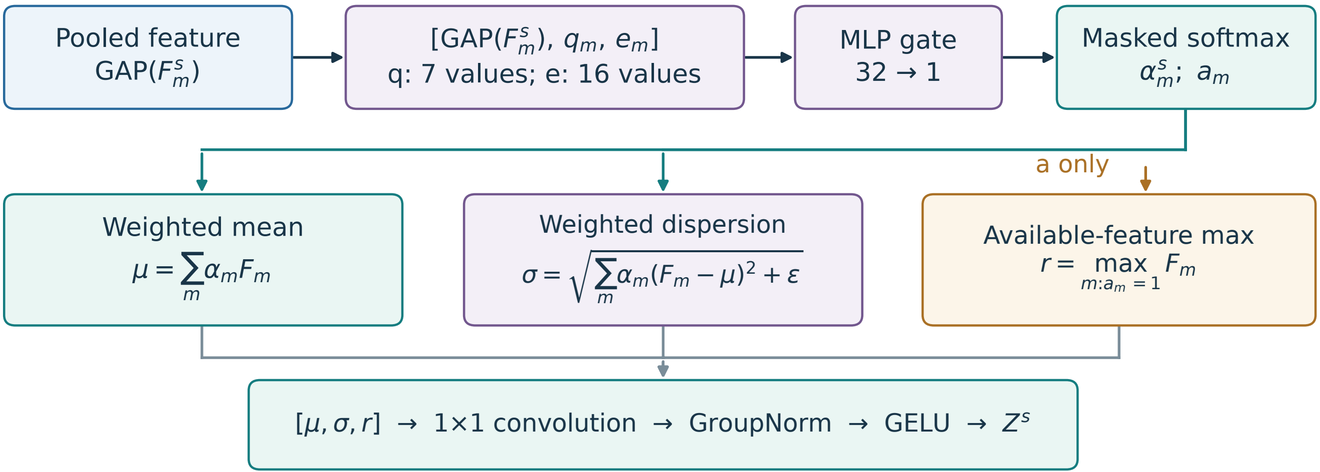
QMMF fusion: descriptors, availability and feature statistics

One independent block per encoder scale; weights are shared across modalities within each gate.

GAP drives the weights; all statistics use the unpooled feature tensors F.



$$\alpha_m = \frac{a_m \exp(\ell_m)}{\sum_j a_j \exp(\ell_j)} \quad \text{with} \quad \ell_m = g([\mathrm{GAP}(F_m), q_m, e_m]) \quad \text{and} \quad \varepsilon = 10^{-6}$$

No quality: omit q only. Shuffled quality: use a different training group's descriptors.
Matched moments: equal available weights, mean + dispersion, widened bottleneck; retains auxiliary heads.